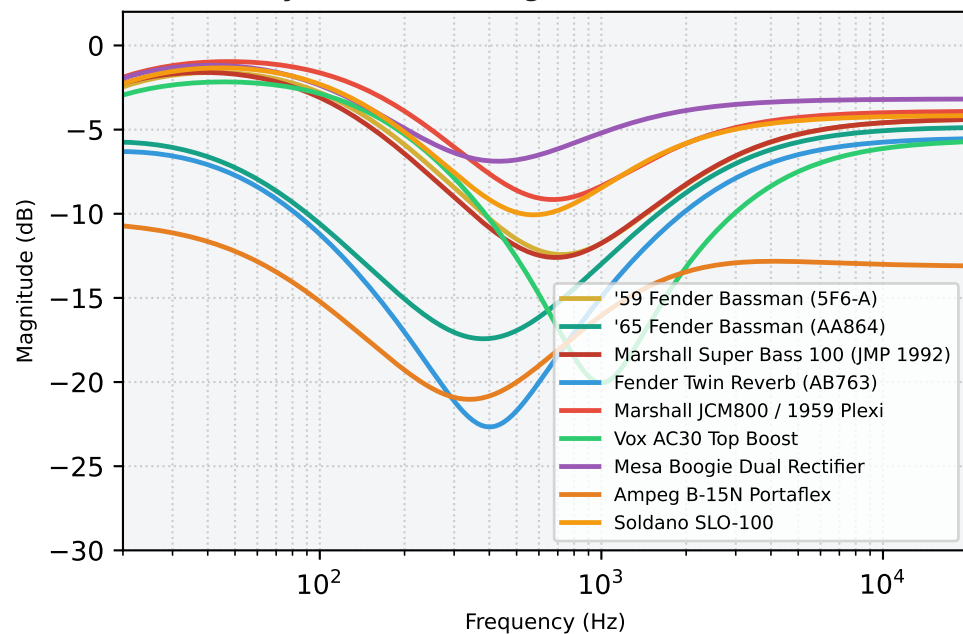


CLASSIC AMP TONE STACK COMPARATIVE BENCHMARK

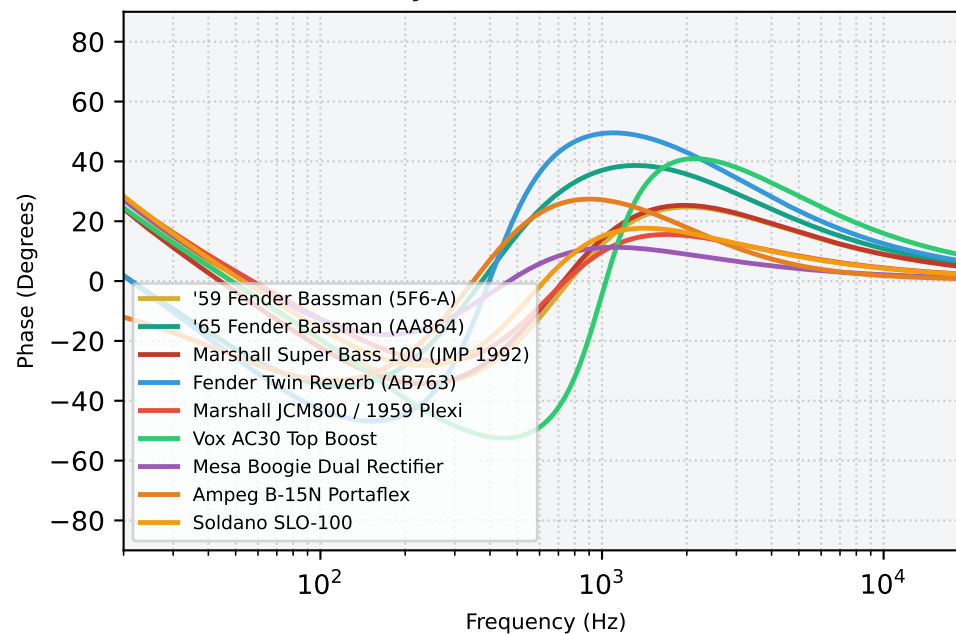
Mathematical Comparison of 6 Iconic Circuit Topologies (Fender, Marshall, Vox, Mesa, Soldano)

Yeh & Smith DAFx-06 Nodal Synthesis

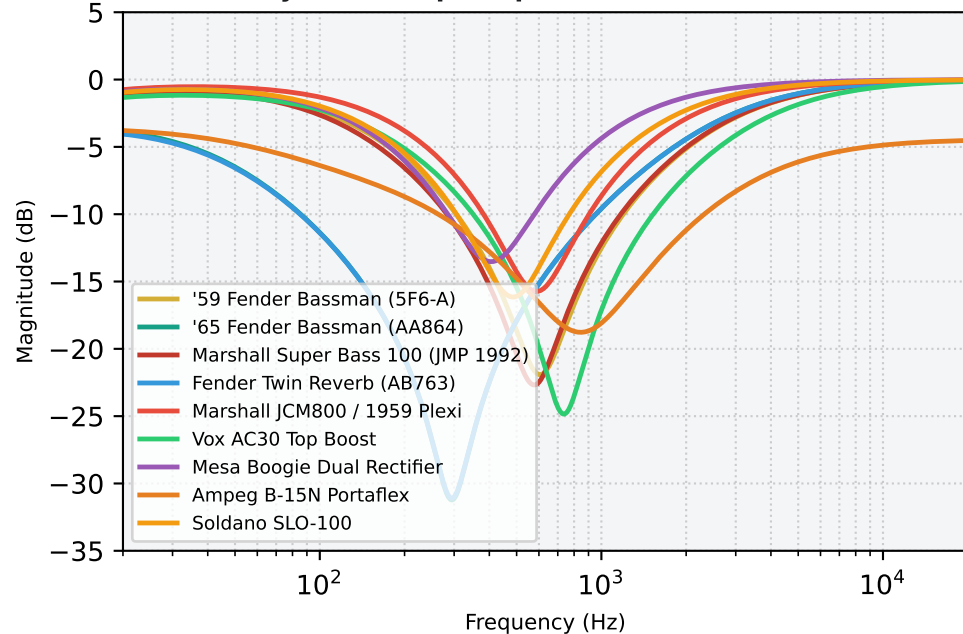
Factory Default Tone Signatures (T=5, M=5, B=5)



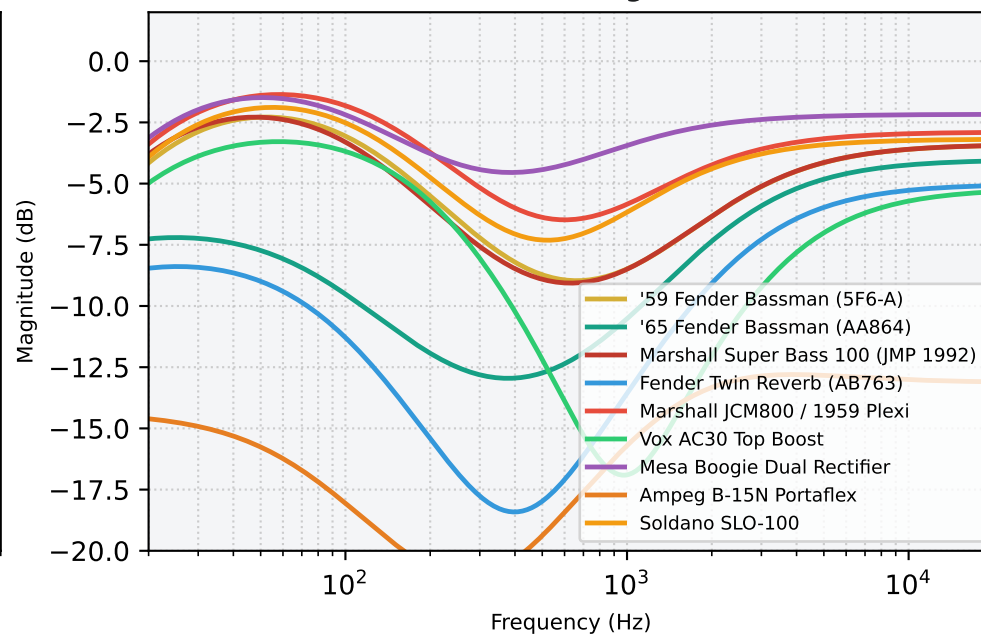
Phase Trajectories across Tone Stacks



Heavy Mid-Scoop Response (T=10, M=0, B=10)



Mid Push / Solo Lead Setting (T=5, M=10, B=3)



Circuit Component Specifications & Acoustic Mapping

Mathematical Component Values and Characteristic Notch Frequencies

Ready for C++ Modular Implementation

Amplifier Circuit		R1 (Treb)	R2 (Bass)	R3 (Mid)	R4 (Slope)	C1 (Treb)	C2 (Bass)	C3 (Mid)	Notch Freq	Acoustic Personality
'59 Fender Bassman (5Y4)	Standard	250k	1.0M	25k	56k	250 pF	20 nF	20 nF	725 Hz	Original Tweed benchmark; balanced warmth, 80dB gain
	Blackface	250k	250k	25k	100k	250 pF	100 nF	47 nF	380 Hz	100nF deep bass cap, 47nF mid cap
Marshall Super Bass 100 (JF)	Standard	250k	1.0M	25k	56k	250 pF	22 nF	22 nF	685 Hz	Heavy tube bass, 56k slope, 250pF treble, 22nF bass cap
Fender Twin Reverb (AE)	Standard	250k	250k	10k	100k	250 pF	100 nF	47 nF	400 Hz	Classic Blackface huge low-end authority, deeper scooped mids
Marshall JCM800 / 1959	Standard	220k	1.0M	22k	33k	470 pF	22 nF	22 nF	672 Hz	British crunch, forward midrange, higher treble bite
Vox AC30 Top Boost	Standard	1000k	1.0M	10k	100k	47 pF	22 nF	10 nF	984 Hz	British chime, 100k slope, small treble cap, 100k slope, sp
Mesa Boogie Dual Rect	Standard	250k	1.0M	50k	47k	680 pF	20 nF	20 nF	629 Hz	Modern metal; 680pF treble cap, 50k mid pot for extreme
Ampeg B-15N Portaf	Standard	470k	1.0M	10k	100k	330 pF	0 nF	22 nF	341 Hz	Legendary tube bass, flat Baxandall stack; flat center, indep
Soldano SLO-100	Standard	250k	1.0M	25k	47k	470 pF	20 nF	20 nF	577 Hz	Boutique high gain; legendary singing lead, smooth